Ligand–Receptor Cross-Validation (LIANA) of Intestinal IEL and LPL Compartments in Crohn's Disease

Bioinformatics Analysis Report

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Contents

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1 Executive Summary

This report presents the LIANA-based ligand–receptor cross-validation of intraepithelial (IEL) and lamina propria (LPL) lymphocyte compartments in Crohn's disease (CD) versus control intestinal tissue, conceived from the outset as an **orthogonal cross-method comparison** for the CellChat v2 findings reported separately (*Cell–Cell Communication Analysis (CellChat v2)…*, companion report). Where CellChat imposes a single mass-action statistical model, LIANA aggregates five conceptually independent scoring methods (CellPhoneDB, NATMI, Connectome, SCA, LogFC) via RobustRankAggreg, deliberately excluding its own internal CellChat reimplementation from the method set to preserve genuine cross-method independence.

Two fully independent LIANA pipelines were run, one per compartment, each following an identical four-stage structure: pooled (non-donor-level) rank aggregation, global exploration, targeted validation of the CellChat pathway deep-dive, and Hallmark over-representation analysis (ORA) on the ligand/receptor gene subset. Across both compartments, the analysis produces three categories of outcome, all reported without selective filtering:

- **Confirmations**: MHC-I (both compartments), CDH1–ITGAE_ITGB7 (IEL), SIRPG–CD47 (LPL) are independently reproduced by LIANA using a fully distinct statistical framework and resource database — the strongest form of validation available in this project.
- **A genuine, three-method-arbitrated disagreement (IL2, LPL)**: LIANA's receptor-complex pattern (high-affinity IL2RA_IL2RB_IL2RG specific to *both* Treg populations) contradicts CellChat's original characterization (high-affinity complex specific to T follicular helper/CD4 TRM-like, not Treg). External literature (constitutive CD25/IL2RA expression on Treg, the textbook basis of Treg-selective IL-2 sensitivity) resolves this in LIANA's favor — and GSEA independently corroborates it: `IL2_STAT5_SIGNALING` is significantly CD-enriched specifically in the **Treg** transcriptome (not in Cytotoxic TRM-like), placing the same pathway in the same population by a third, complementary ligand–receptor scoring methods (Section 7). GSEA and LIANA now agree with each other and with canonical immunology; CellChat's receptor-complex assignment is the outlier on all three counts — a rare and instructive case where two orthogonal methods jointly outperform the primary one, reported as such rather than smoothed over.
- **A case where cross-validation caught and corrected a primary-analysis error (MHC-II, LPL)**: LIANA's finding that Activated effector Treg receives substantial MHC-II signal in CD directly contradicted the original CellChat findings-log claim of "0 interactions in CD" for this population. Rather than recording this as an unresolved LIANA–CellChat divergence, the discrepancy was traced back to the CellChat object itself via `subsetCommunication()`, which showed the original claim was

not supported by CellChat's own output (37 significant CD interactions vs 28 in Control, comparable magnitude). The correction — "MHC-II repertoire broadening in CD," not "population substitution" — has already been propagated into the CellChat findings log and the CellChat final report (Sections 6.2 and 10, companion report). This is the single most concrete demonstration in this project of why an orthogonal cross-method comparison was built in the first place.

Two further points define the boundaries of this analysis. First, an unresolved tension worth flagging for the manuscript: for several targeted pathway validations, the naive top-N ranking approach that had worked cleanly for IEL's exploratory dotplots proved unsuitable for LPL's pathway-specific cross-validation questions, because ubiquitous, high-magnitude signals (generic MHC-I self-recognition, adhesion) systematically outranked the weaker, biologically targeted pathways under test — resolved by switching to direct gene-level filtering rather than dotplot ranking for cross-validation questions specifically (Section 8). Second, GSEA cross-validation (Section 7) is complete for IEL and LPL compartments.

This report **does not introduce new computation**; it consolidates the four analyst-produced documents (`LIANA_IEL_findings_log.md`, `LIANA_IEL_pipeline_report.md`, `LIANA_LPL_findings_log.md`, `LIANA_LPL_pipeline_report.md`) into a single, publication-oriented narrative, structured identically to the CellChat final report for direct cross-referencing.

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2 Analytical Rationale

2.1 Why LIANA as an orthogonal cross-method comparison

CellChat imposes a single statistical model (law-of-mass-action communication probability, permutation test) and a single curated database (`CellChatDB`). Any finding that depends on modeling choices specific to that combination cannot, by construction, be distinguished from a genuine biological signal using CellChat alone. LIANA addresses this by combining five methods with different statistical bases (permutation-based, edge-specificity-based, log-fold-change-based) over an independently curated resource (`Consensus`, ~4700 interactions), explicitly excluding its internal CellChat reimplementation from the aggregated method set. A finding that survives both frameworks is substantially more defensible than one supported by either alone; a finding that does not survive is either a modeling artifact worth flagging (as with IL2) or, in one case, evidence that the primary finding itself needs correction (MHC-II).

2.2 Pipeline architecture

- Two fully independent pipelines, one per compartment, each built as: manual `SingleCellExperiment` construction (bypassing a Seurat v5/`GetAssayData()` incompatibility — Section 8) → `liana_wrap(method = c("natmi","connectome","logfc","sca","cellphonedb"), resource = "Consensus")` → `rank_aggregate()` → global exploration (`heat_freq`, `chord_freq`) → targeted validation against the CellChat pathway deep-dive → `liana_bysample(sample_col = "disease")` for the CD-vs-Control split → Hallmark ORA on ligand and receptor gene subsets.
- **Design choice, not a limitation**: both compartments use a pooled, non-donor-level architecture. Verified against primary source (sc-best-practices.org, Cell–Cell Communication chapter): the underlying scoring methods were designed for steady-state, single-condition data; the tensor-based multi-condition extension (`tensor_cell2cell`) is an explicitly optional module, not the base workflow, and was not used — it also would not accommodate this dataset's limited, shared donor structure across both compartments.
- Reproducibility: `set_seed(1234)` called identically in both compartments, consistent with the CellChat pipeline's seeding convention.

2.3 Scope and a key methodological asymmetry vs. CellChat

Both compartments use a `pmin(CD, Control) < 30` threshold to define the "fully comparable" tier for CD-vs-Control dotplots and ORA — identical logic and threshold in IEL and LPL, but **not identical to CellChat's own validity tables** for the same compartments (CellChat: `filterCommunication(min.cells = 10)`, applied per-condition rather than as a joint minimum). This is most consequential in LPL, where the LIANA "fully comparable" set (7 populations) is substantially broader than CellChat's own "fully comparable" set (4 populations, Section 3.2 of the companion CellChat report). This is not an inconsistency to silently reconcile — it is declared explicitly in Section 3 below, and readers should not assume the two reports' "fully comparable" labels refer to the same population set.

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3 Methodological Framework: Cell Type Comparison Validity (LIANA-specific)

3.1 IEL compartment

Tier	Cell types (n CD / n Control)	Interpretation
Fully comparable (`min(CD,Control)  30`)	CXCL13 TFH-like T cells (75/1394), GZMK effector memory CD8 T cells (183/686), Effector T cell mixed state (213/356), Cycling T cells (207/35)	Retained for the primary CD-vs-Control comparison
Imbalanced	CD39 tissue-resident CD8 T cells (2551/20), Cytotoxic T cells (2113/11), NK-like cytotoxic T cells (249/10), CST3 LYZ macrophages (35/1), TH17 effector (4/1508), others with `min < 30`	Comparison possible; weak side indicative only
Condition-exclusive / quasi-exclusive	FOXP3 IL2RA Treg (79/0), Terminal effector CD8 T cells (529/0), IL7R memory-like CD8 T cells (0/1481)	Within-condition characterization only

This table is identical in logic and, for IEL, in resulting population set to the CellChat IEL validity table (companion report, Section 3.1) — both compartments' underlying cell composition tables are shared, and IEL's imbalance structure happens to produce the same practical grouping under both criteria.

3.2 LPL compartment

Tier	Cell types (n CD / n Control)	Interpretation
Fully comparable (LIANA threshold, `min(CD,Control)` `30`)	Activated cytotoxic CD8 T cells (5650/2002), Naive CD4 T cells (2284/3756), Activated effector Treg (873/617), CD4 TRM-like activated memory T cells (753/249), Th17-like activated CD4 T cells, IFN-activated CD8 cytotoxic T cells, T cells / innate cytotoxic T cells	**Broader than CellChat's own "fully comparable" set** (4 populations, companion report Section 3.2) — different validity criterion, declared explicitly, not to be treated as the same set
Imbalanced	Th17-like activated CD4 T cells (Control n=53), IFN-activated CD8 cytotoxic T cells (Control n=47), T cells / innate cytotoxic T cells (Control n=57), Innate-like / TRM CD8 (2194 CD/16 Control), **Cycling T cells**	**Cycling T cells falls below the LIANA <30 threshold** despite being treated as "well represented in both conditions" in the CellChat SIRP deep-dive — a direct instance of the two tools' validity criteria disagreeing on a single population; declared rather than reconciled
Condition-exclusive	FOXP3 activated Treg (n=0 Control), NK-like cytotoxic T cells (n=0 Control), TRM-like CD8 cytotoxic T cells (n=0 Control)	Within-condition characterization only; appearance of these populations in pooled or CD-only results must always be checked for compositional tautology before interpretation

Structural note (shared with CellChat). LPL contains no myeloid or epithelial populations; this is a genuine structural feature of the compartment, not an analytical artifact, and constrains the LIANA global exploration design identically to how it constrains CellChat's (Section 3.2, companion report).

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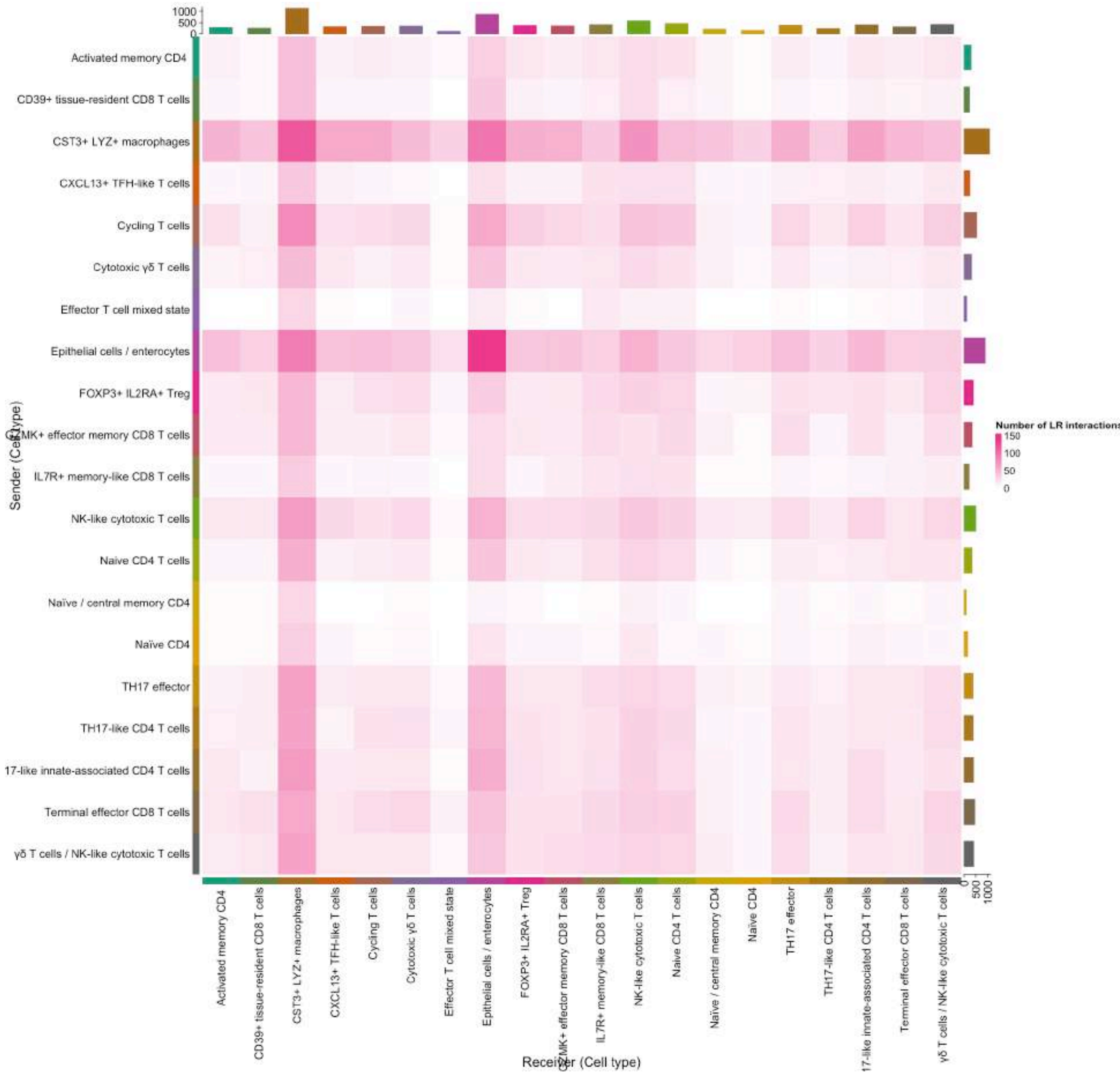
4 Global Network Architecture

	IEL	LPL
`heat_freq` dominant receivers (pooled, `sp ecificity_rank` `0.05`)	CST3 LYZ macrophages, Epithelial cells — consistent with expected myeloid/epithelial structural role	No myeloid/epithelial hub (none present); diffuse T–T communication pattern
`chord_freq` design	Reused CellChat's `celltypes_of_interest` set for consistency (mixed lineage)	Redefined on a T–T-only basis (no mixed-lineage analogue available)

	IEL	LPL
`chord_freq` notable result	Not specifically flagged as informative beyond confirming global structure	Group 2 (Naive CD4 T cells, FOXP3 activated Treg, Th17-like activated CD4 T cells): **Naive CD4 T cells appears visually isolated** after the joint `specificity_rank/magnitude_rank` filter — an independent LIANA-sourced confirmation of CellChat's characterization of this population as having no dominant signal on any tested pathway

Figure 1a.



Sender (Cell type)
Receiver (Cell type)
Number of LR interactions
150
100
50
0
1000
500
0
Activated memory CD4
CD39+ tissue-resident CD8 T cells
CST3+ LYZ+ macrophages
CXCL13+ TFH-like T cells
Cycling T cells
Cytotoxic γδ T cells
Effector T cell mixed state
Epithelial cells / enterocytes
FOXP3+ IL2RA+ Treg
GZMK+ effector memory CD8 T cells
IL7R+ memory-like CD8 T cells
NK-like cytotoxic T cells
Naive CD4 T cells
Naïve / central memory CD4
Naïve CD4
TH17 effector
TH17-like CD4 T cells
17-like innate-associated CD4 T cells
Terminal effector CD8 T cells
γδ T cells / NK-like cytotoxic T cells

Figure 1b.

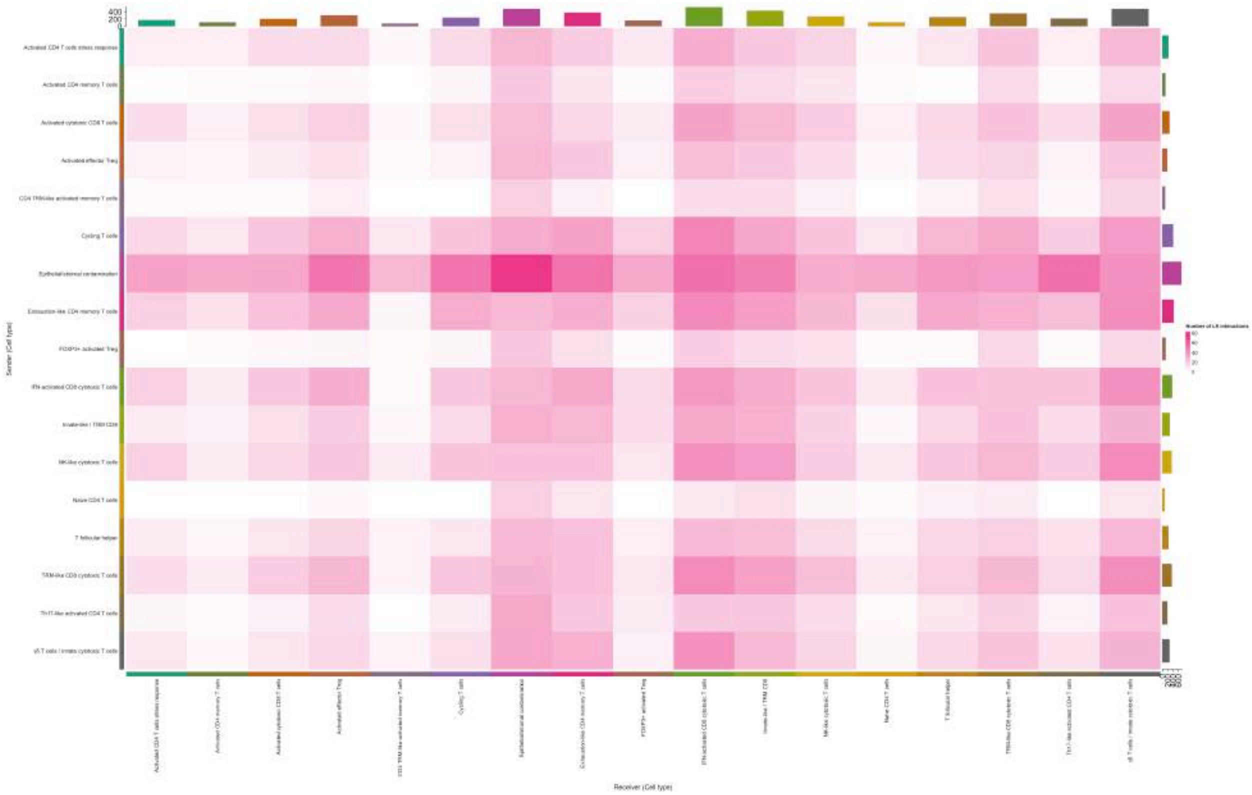
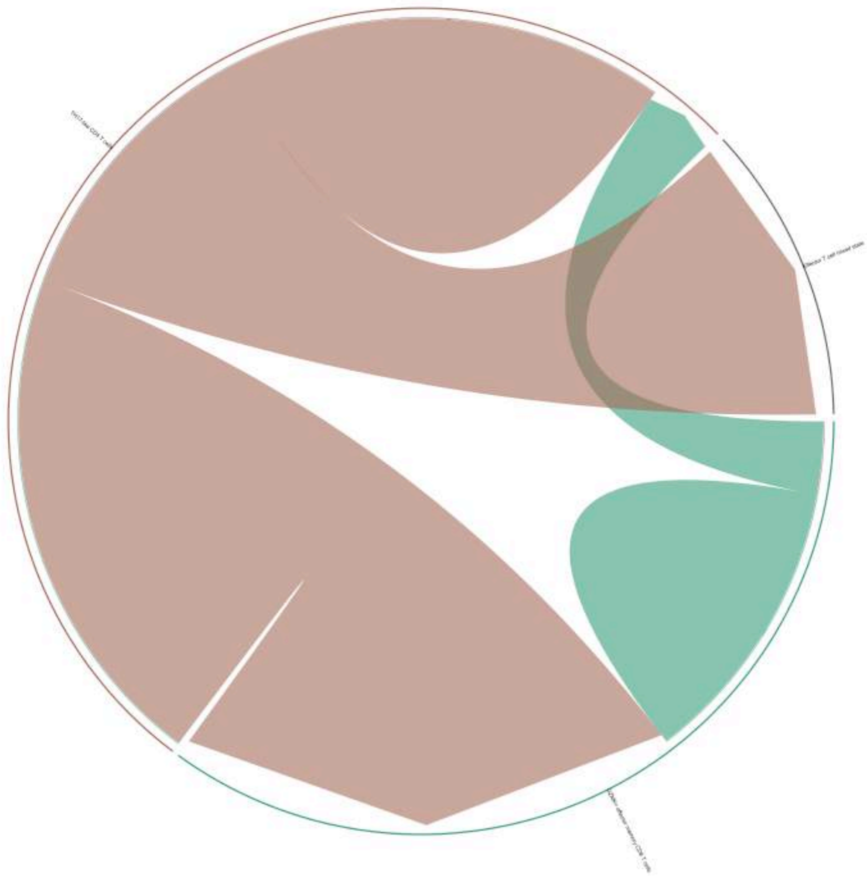


Figure 2a.



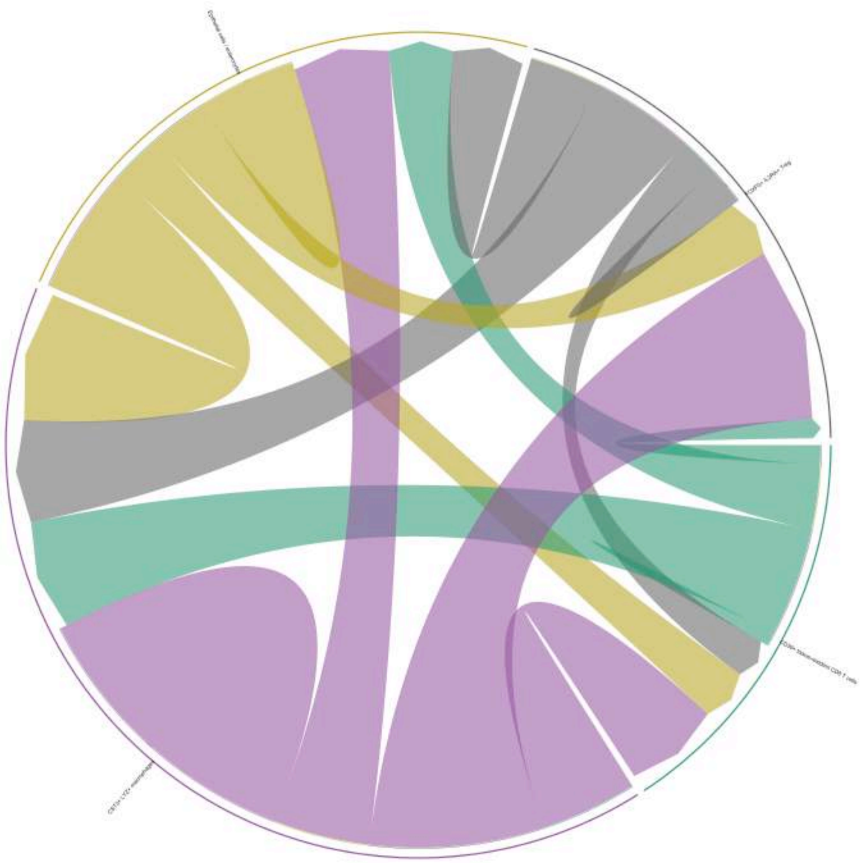
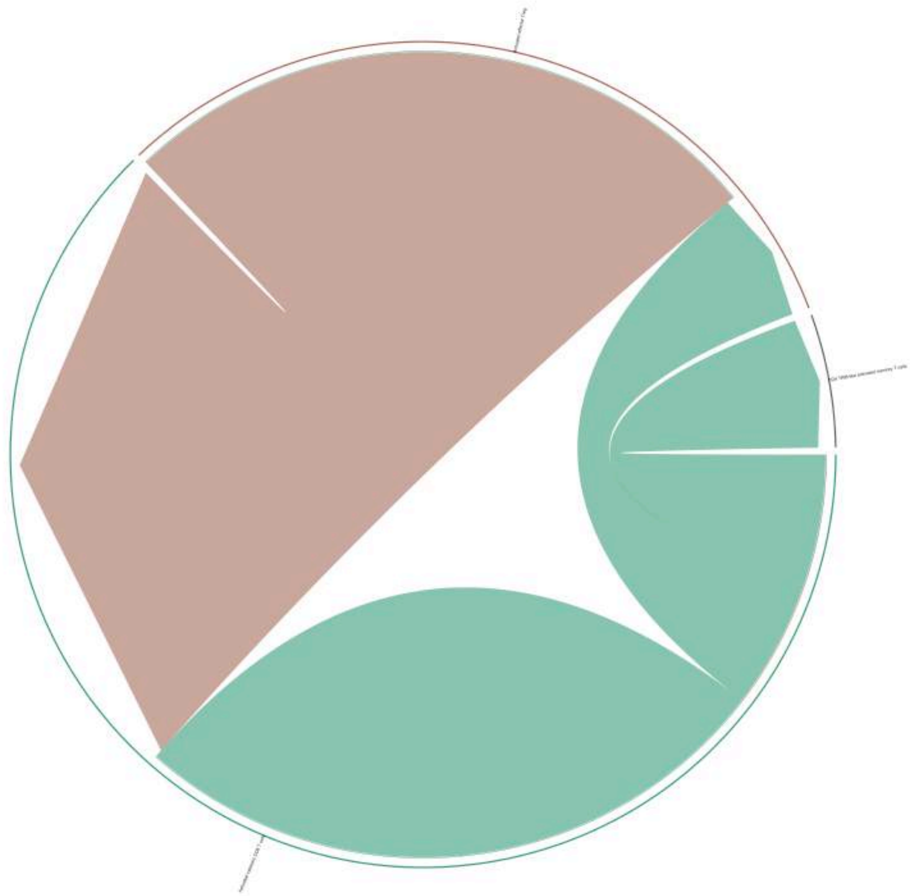
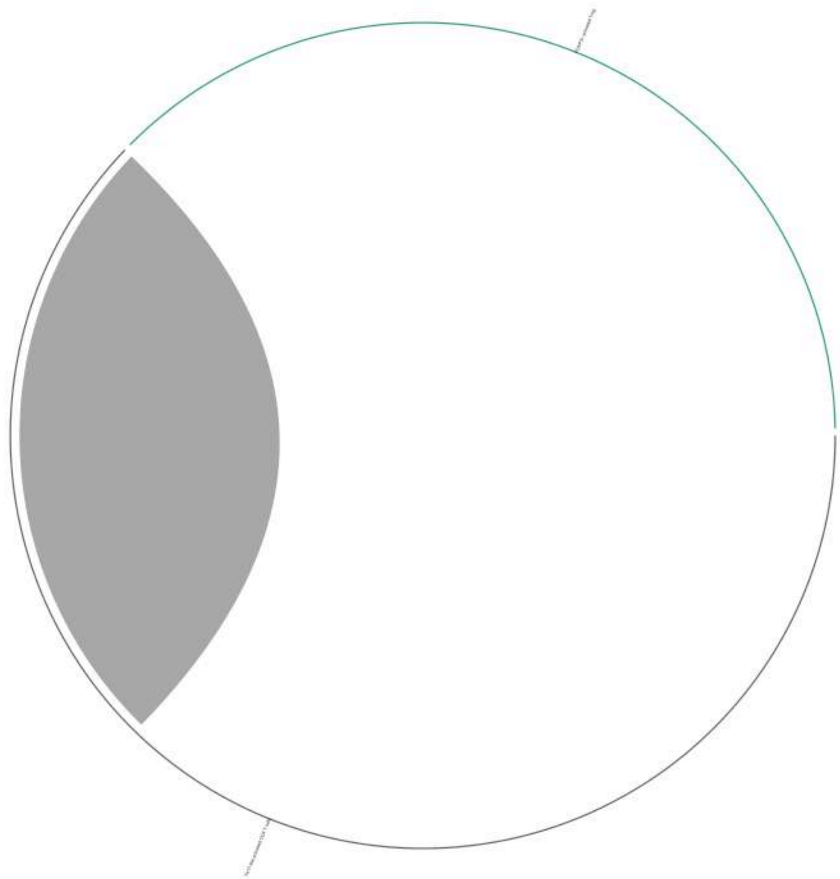


Figure 2b.





5 IEL Compartment: Validation, CD-vs-Control, and Enrichment Findings

5.1 Targeted validation of the CellChat pathway deep-dive

CellChat finding	LIANA outcome	Evidence
MHC-I (valid without caveat in CellChat)	**Confirmed independently**, no targeted filter needed	Dominates the unfiltered top-20 ranking and the `heat_freq` overview
CDH1 → ITGAE_ITGB7 (most specific CellChat finding)	**Confirmed**, high specificity	`spe cificity_rank = 0.0048` (top ~0.5%); `magnitude_rank = 0.499` explains its absence from magnitude-ranked views despite being highly specific

CellChat finding	LIANA outcome	Evidence
MHC-II (CellChat: population substitution, TH17 effector → FOXP3 Treg)	Signal present but **confounded by compositional constraint**, not independently testable	Treg (79 CD/0 Control) and TH17 effector (4 CD/1508 Control) are near-mutually-exclusive by construction — the CD-vs-Control receiver pattern in `liana_bysample()` is an arithmetic consequence of composition, not an independent finding
LIGHT (TNFSF14) → LTBR ("within-CD" in CellChat)	**Not testable**, cause diagnosed with certainty	Pair present in Consensus resource; TNFSF14 well-expressed in sender (20.6% CD39 TRM); **LTBR at 8.3% in CST3 LYZ macrophages, just below the default `expr_prop = 0.10` threshold** (1.7-point margin); re-tested on CD-only data still absent, making a pooling-related artifact unlikely

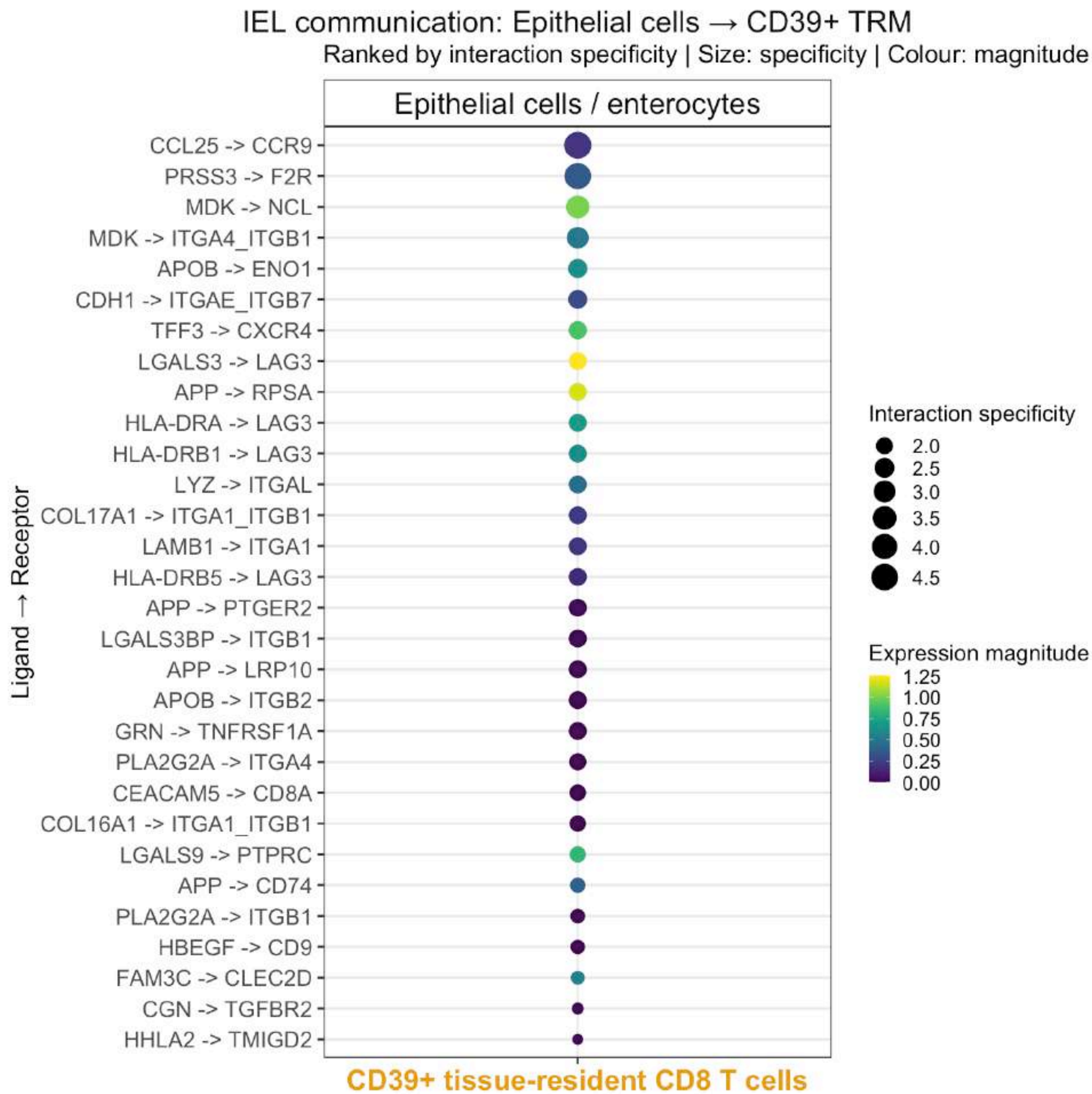
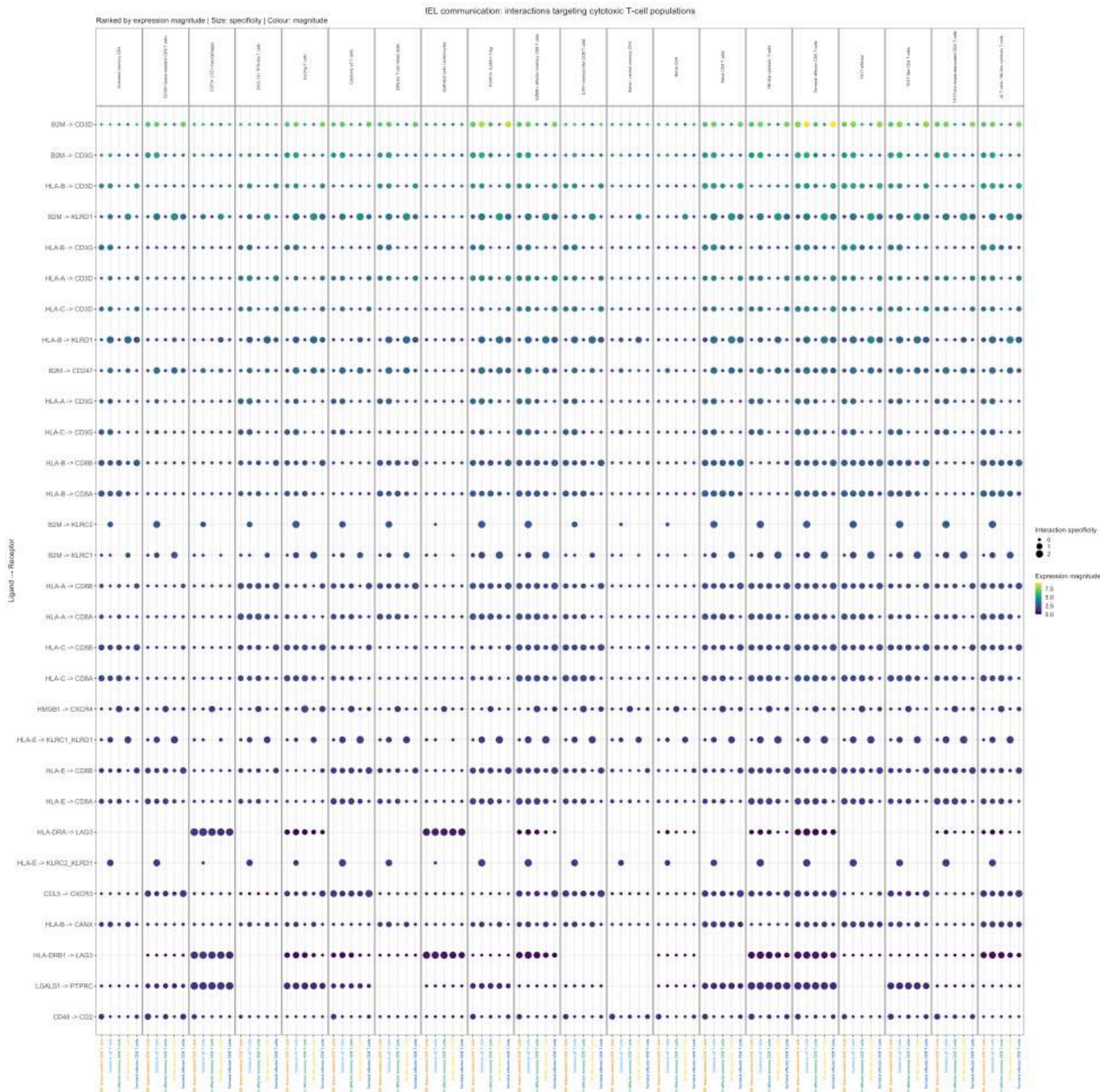
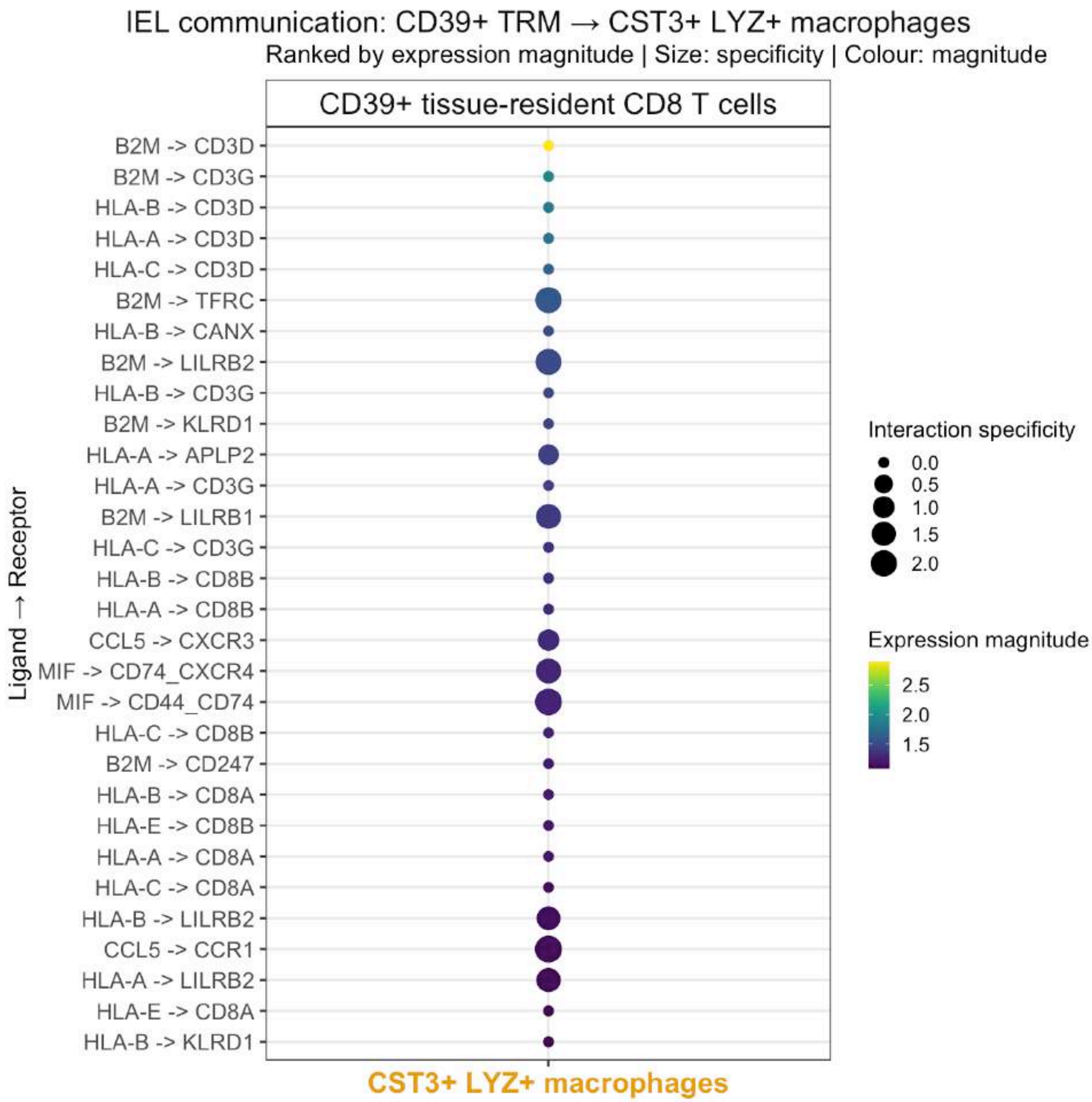

IEL communication: Epithelial cells → CD39+ TRM
Ranked by interaction specificity | Size: specificity | Colour: magnitude
Epithelial cells / enterocytes
CCL25 -> CCR9
PRSS3 -> F2R
MDK -> NCL
MDK -> ITGA4_ITGB1
APOB -> ENO1
CDH1 -> ITGAE_ITGB7
TFF3 -> CXCR4
LGALS3 -> LAG3
APP -> RPSA
HLA-DRA -> LAG3
HLA-DRB1 -> LAG3
LYZ -> ITGAL
COL17A1 -> ITGA1_ITGB1
LAMB1 -> ITGA1
HLA-DRB5 -> LAG3
APP -> PTGER2
LGALS3BP -> ITGB1
APP -> LRP10
APOB -> ITGB2
GRN -> TNFRSF1A
PLA2G2A -> ITGA4
CEACAM5 -> CD8A
COL16A1 -> ITGA1_ITGB1
LGALS9 -> PTPRC
APP -> CD74
PLA2G2A -> ITGB1
HBEGF -> CD9
FAM3C -> CLEC2D
CGN -> TGFBR2
HHLA2 -> TMIGD2
Ligand → Receptor
CD39+ tissue-resident CD8 T cells
Interaction specificity
2.0
2.5
3.0
3.5
4.0
4.5
Expression magnitude
1.25
1.00
0.75
0.50
0.25
0.00

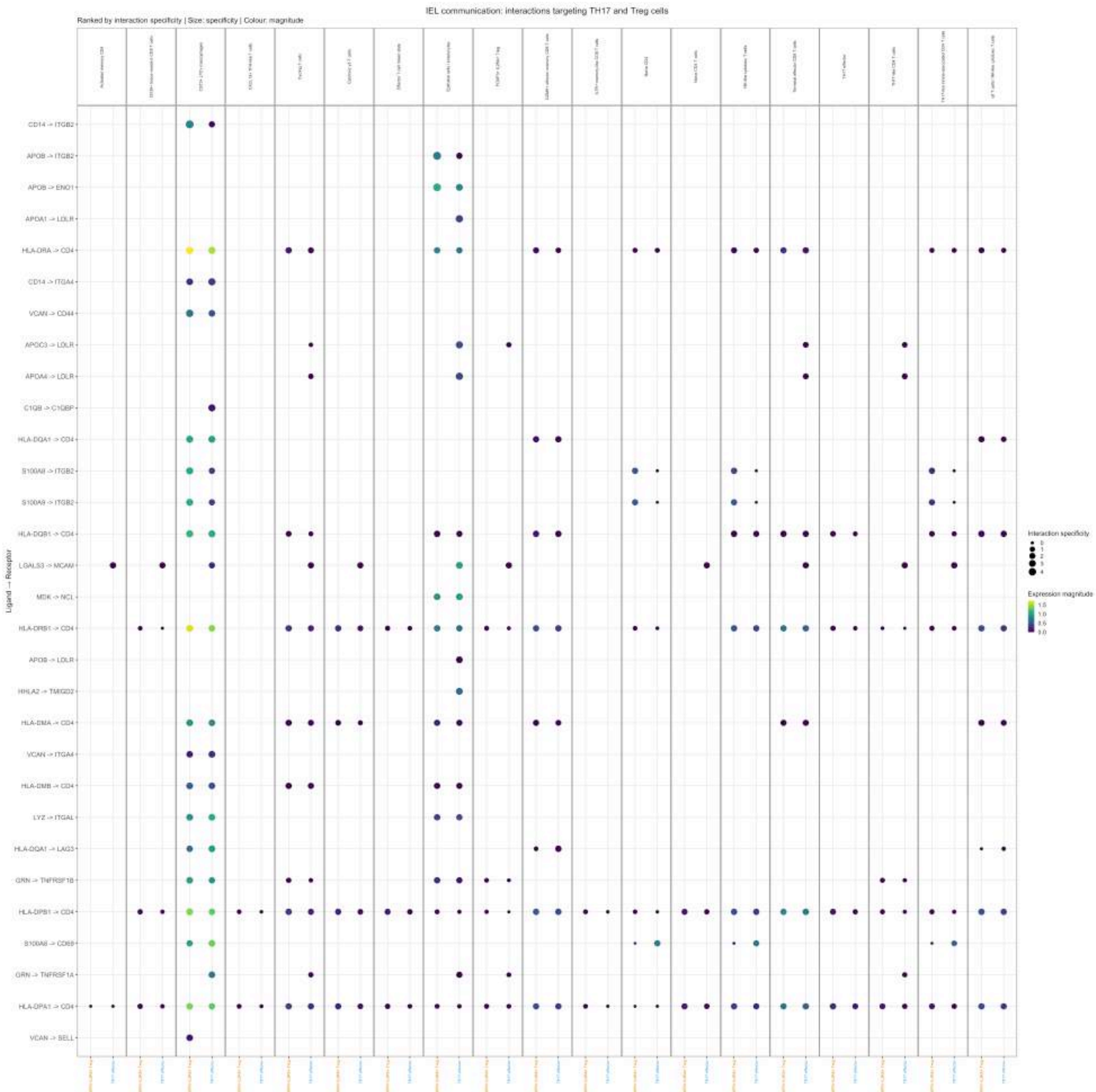
Figure 3.



IEL communication: interactions targeting cytotoxic T-cell populations
Ranked by expression magnitude | Size: specificity | Colour: magnitude
Ligand → Receptor
B2M -> CD3D
B2M -> CD3G
HLA-B -> CD3D
B2M -> KLRD1
HLA-B -> CD3G
HLA-A -> CD3D
HLA-C -> CD3D
HLA-B -> KLRD1
B2M -> CD247
HLA-A -> CD3G
HLA-C -> CD3G
HLA-B -> CD8B
HLA-B -> CD8A
B2M -> KLRC2
B2M -> KLRC1
HLA-A -> CD8B
HLA-A -> CD8A
HLA-C -> CD8B
HLA-C -> CD8A
HMGB1 -> CXCR4
HLA-E -> KLRC1_KLRD1
HLA-E -> CD8B
HLA-E -> CD8A
HLA-DRA -> LAG3
HLA-E -> KLRC2_KLRD1
CCL5 -> CXCR3
HLA-B -> CANX
HLA-DRB1 -> LAG3
LGALS1 -> PTPRC
CD48 -> CD2
Interaction specificity
0
1
2
Expression magnitude
7.5
5.0
2.5
0.0



IEL communication: CD39+ TRM → CST3+ LYZ+ macrophages
Ranked by expression magnitude | Size: specificity | Colour: magnitude
CD39+ tissue-resident CD8 T cells
B2M -> CD3D
B2M -> CD3G
HLA-B -> CD3D
HLA-A -> CD3D
HLA-C -> CD3D
B2M -> TFRC
HLA-B -> CANX
B2M -> LILRB2
HLA-B -> CD3G
B2M -> KLRD1
HLA-A -> APLP2
HLA-A -> CD3G
B2M -> LILRB1
HLA-C -> CD3G
HLA-B -> CD8B
HLA-A -> CD8B
CCL5 -> CXCR3
MIF -> CD74_CXCR4
MIF -> CD44_CD74
HLA-C -> CD8B
B2M -> CD247
HLA-B -> CD8A
HLA-E -> CD8B
HLA-A -> CD8A
HLA-C -> CD8A
HLA-B -> LILRB2
CCL5 -> CCR1
HLA-A -> LILRB2
HLA-E -> CD8A
HLA-B -> KLRD1
Ligand → Receptor
CST3+ LYZ+ macrophages
Interaction specificity
0.0
0.5
1.0
1.5
2.0
Expression magnitude
2.5
2.0
1.5

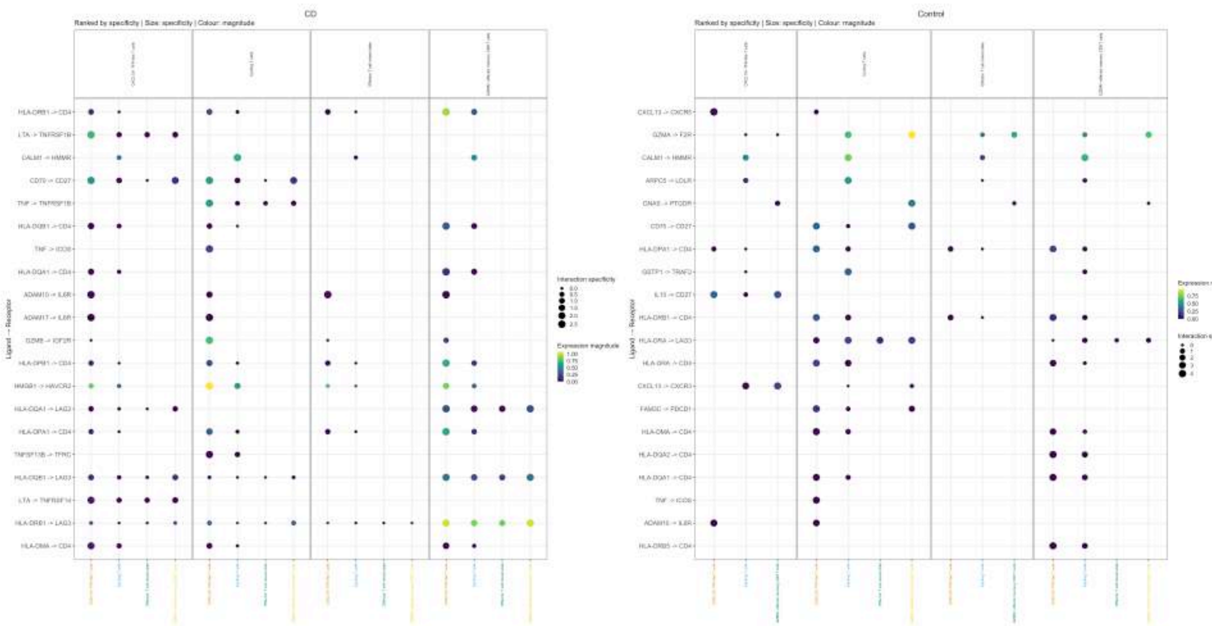


5.2 CD-vs-Control comparison, three levels of granularity

1. **Cytotoxic targets (MHC-I, magnitude-ranked)**: structure preserved between CD/Control, consistent with CellChat's characterization of MHC-I as structural rather than condition-dependent — though the target set includes imbalanced/condition-exclusive populations and should be read with that caveat.
2. **TH17 effector / Treg (specificity-ranked)**: **tautological**, not independent evidence — re-confirmed via `table(celltype, disease)`, the identical constraint already documented for CellChat.
3. **Fully comparable set (4×4, both sides restricted) — the single most defensible CD-vs-Control comparison in the IEL analysis**: a **TNF-superfamily axis active in CD** (`LTA→TNFRSF1B`, `TNF→TNFRSF1B/ICOS`, `TNFSF13B→TFRC`, `ADAM10/17→IL6R`, `GZMB→IGF2R`) against a **regulatory/homeostatic axis in Control** (`IL10→CD27`, `FAM3C→PDCD1` checkpoint, `CXCL13→CXCR5/CXCR3`

follicular homing). No CD-emergent populations involved — free of the compositional confounding affecting (2).

Figure 4.

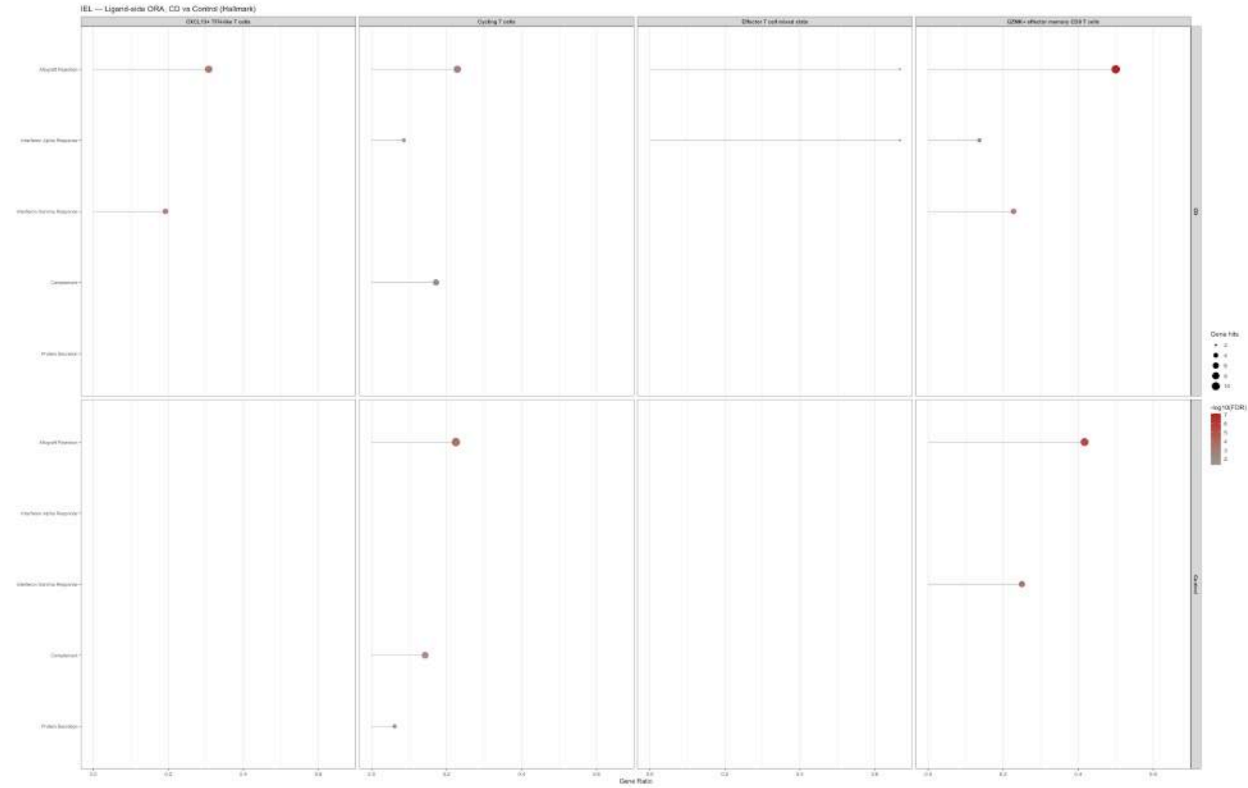


5.3 Hallmark ORA (ligand and receptor side, `fully_comparable_types`)

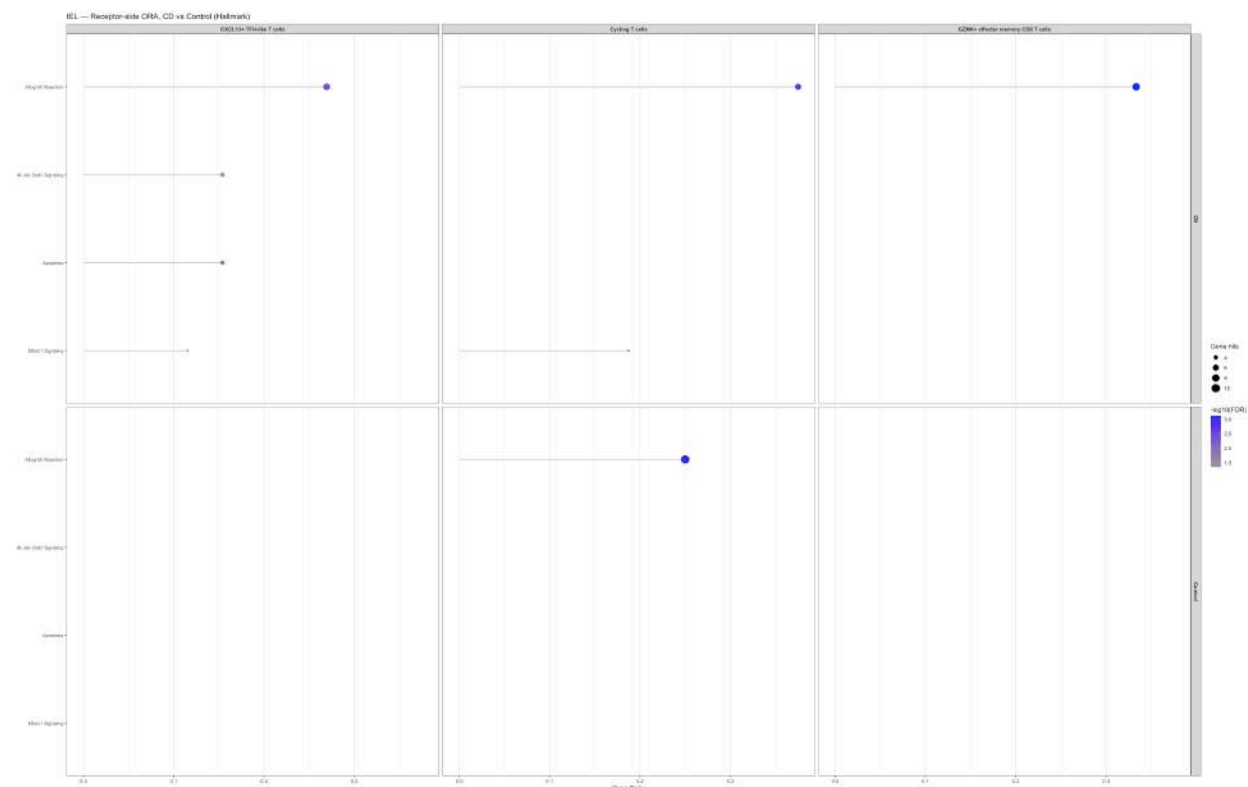
- **Strongest finding**: CXCL13 TFH-like T cells, receptor side — **4 pathways exclusive to CD**, none in Control: `IL6_JAK_STAT3_SIGNALING`, `APOPTOSIS`, `MTORC1_SIGNALING`, `ALLOGRAFT_REJECTION`. IL6/JAK/STAT3 is a validated Crohn's disease therapeutic target (tocilizumab, JAK inhibitors) — clinical concordance independent of the analysis design.
- **Declared caution**: `ALLOGRAFT_REJECTION` recurs across nearly every cell type tested — a broad, immune-centric gene set, expected given a background already restricted to immune ligand/receptor genes; treated throughout as "expected/control," never as a discriminating finding on its own.
- **Fragile hit, quantified**: Effector T cell mixed state, ligand side, CD, `ALLOGRAFT_REJECTION` — `ligands_in_gs=2`, `distinct_hits=3`, `adj_pval=0.0475` (just below threshold) — reported as weak, not on par with the four CD-exclusive hits above.
- **Genuine negative result**: Effector T cell mixed state entirely absent from the receptor side (no Hallmark pathway above threshold in either condition) — tested and reported, not omitted.

Figure 5.

Ligand


IEL — Ligand-side ORA, CD vs Control (Hallmark)
Gene Ratio

Receptor


IEL — Receptor-side ORA, CD vs Control (Hallmark)
Gene Ratio

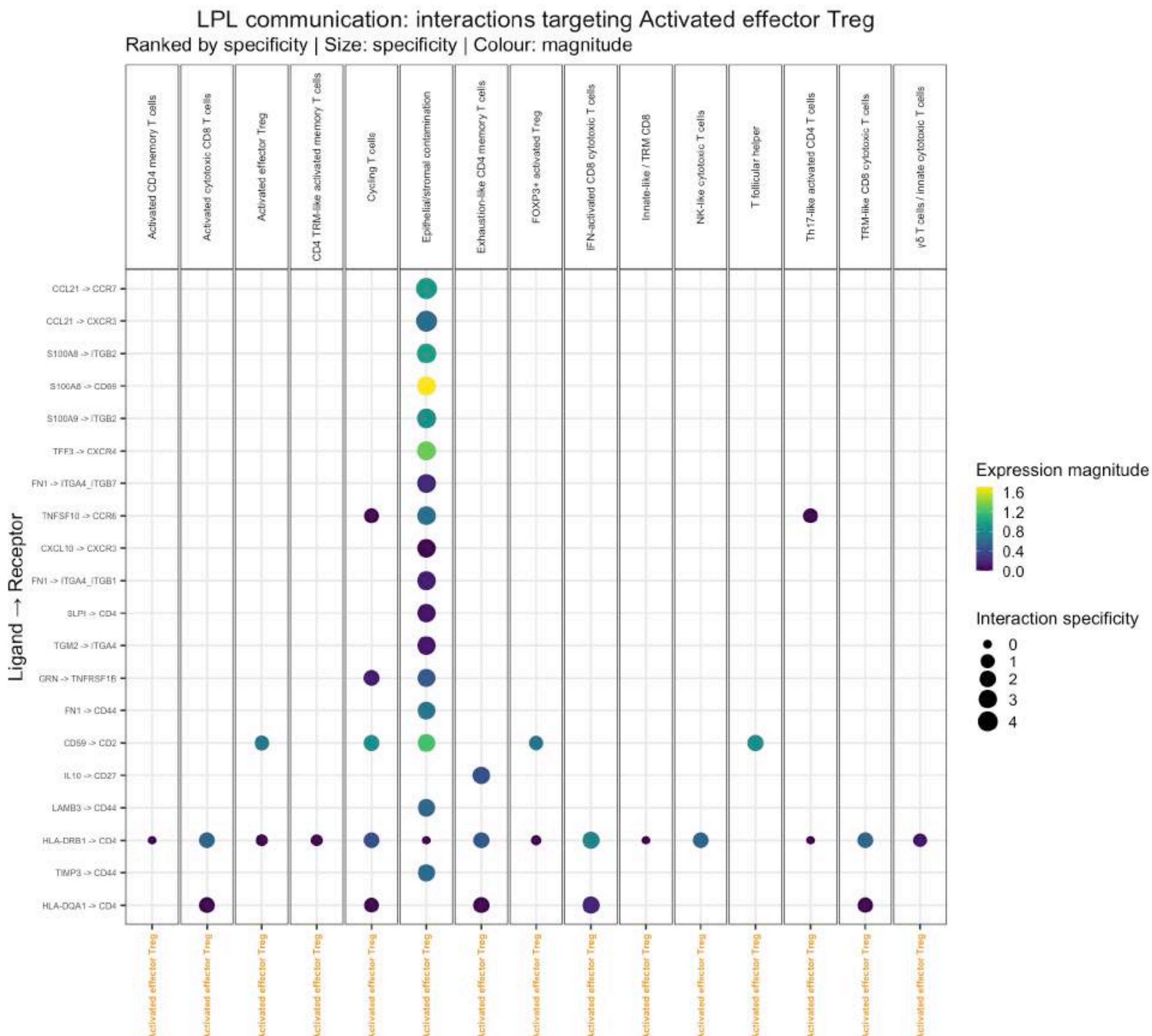
6 LPL Compartment: Validation, CD-vs-Control, and Enrichment Findings

6.1 Targeted validation of the CellChat pathway deep-dive

CellChat finding	LIANA outcome	Evidence
MHC-I on Activated effector Treg (strongest quantitative LPL finding)	**Confirmed**, with a necessary distinction	Canonical HLA-A/B/C→CD8A/CD8B pairs present but `specificity_rank` near 1 (non-specific, ubiquitous) — consistent with CellChat's own "broad, near-ubiquitous" characterization. The unfiltered top-30 ranking is instead dominated by **B2M→TFRC** and **HLA-A→APLP2** — verified via `select _resource("Consensus")` as genuinely curated (CellPhoneDB + connectomeDB2020) but biologically distinct: the B2M–HFE–TFRC iron-homeostasis axis, not MHC-I antigen presentation. The MHC-I claim must be anchored to the HLA-*→CD8A/CD8B rows specifically, never to B2M→TFRC/APLP2.

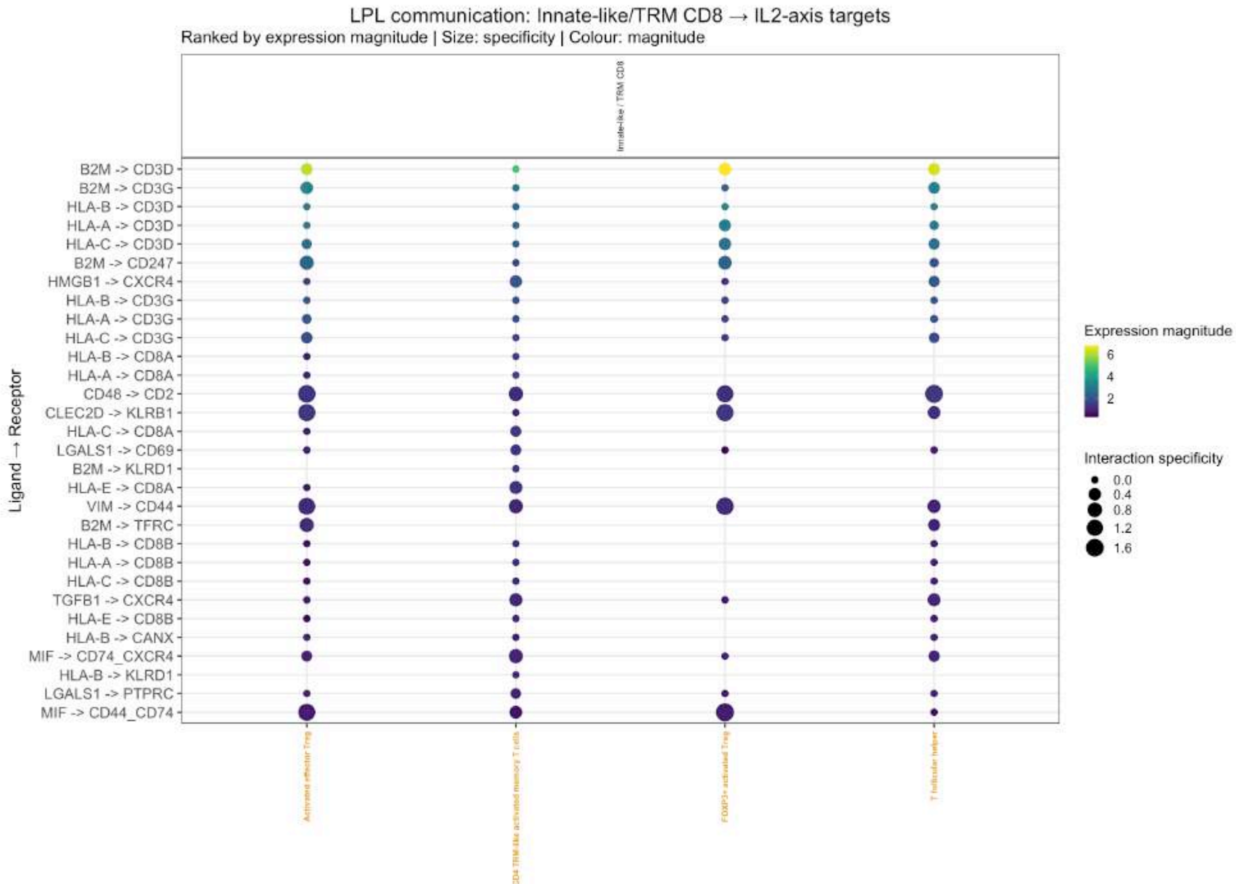
CellChat finding	LIANA outcome	Evidence
IL2 (CellChat: high-affinity IL2RA–IL2RB specific to T follicular helper/CD4 TRM-like, not Treg)	**Disagreement, resolved in LIANA's favor by external literature**	LIANA shows the opposite pattern: the full high-affinity trimeric complex **IL2RA__IL2RB__IL2RG** is specific to **both** Treg populations (Activated effector Treg and FOXP3 activated Treg, `spec ificity_rank = 0.00802` in both), while T follicular helper/CD4 TRM-like show only the intermediate-affinity IL2RB_IL2RG/CD53. Literature verification: Treg constitutively express CD25/IL2RA, the textbook basis of the Treg-selective high-affinity trimeric receptor — the LIANA pattern is the biologically canonical one; CellChat's is inverted relative to established immunology, plausibly reflecting receptor-complex annotation differences between `CellChatDB` and `Consensus`. Additionally, CellChat's "0 interactions in Control" claim (CD-exclusive) is better stated, per LIANA, as "markedly attenuated but not absent" (`magnitude_rank` near 1 in Control vs 0.4–0.6 in CD) rather than strictly absent.
SIRPG–CD47 (CellChat: network reorganization among Cycling T cells, CD4 TRM-like, Activated effector Treg)	**Confirmed**, with an additional supporting detail	Pair present in both conditions among the three core populations. In CD, CD4 TRM-like activated memory T cells does not act as a sender (only Activated effector Treg and Cycling T cells send), while all three send in Control — a second, independent line of evidence for CellChat's "reorganization, not simple loss" reading (rankSimilarity), not merely a magnitude reduction.

CellChat finding	LIANA outcome	Evidence
MHC-II (CellChat original claim: population substitution, Activated effector Treg in Control → FOXP3 activated Treg in CD)	**Original CellChat claim found unsupported by CellChat's own object** — **corrected**, not a LIANA-CellChat divergence	LIANA showed substantial, significant MHC-II signal toward Activated effector Treg in CD (multiple HLA-DRB1/DPB1/DPA1/DQB1→CD4 rows, `specificity_rank` 0.005–0.02), contradicting "0 interactions in CD." Direct verification via `subsetCommunication(cellchat_lpl_ cd/ctrl, signaling="MHC -II", targets.use="Activated effector Treg")`: **confirmed Activated effector Treg receives MHC-II in both conditions** — CD: 37 significant interactions (`pval=0`) from 9 senders; Control: 28 from 6 senders; comparable magnitude order between conditions. Part of the sender-side CD expansion is tautological (two of three new CD senders are CD-exclusive populations by construction); the sole non-tautological addition is autocrine signaling (Treg→itself), CD-only. **Corrected reading**: repertoire broadening, not population substitution — already propagated to the CellChat findings log and final report (Sections 6.2, 10, companion report).

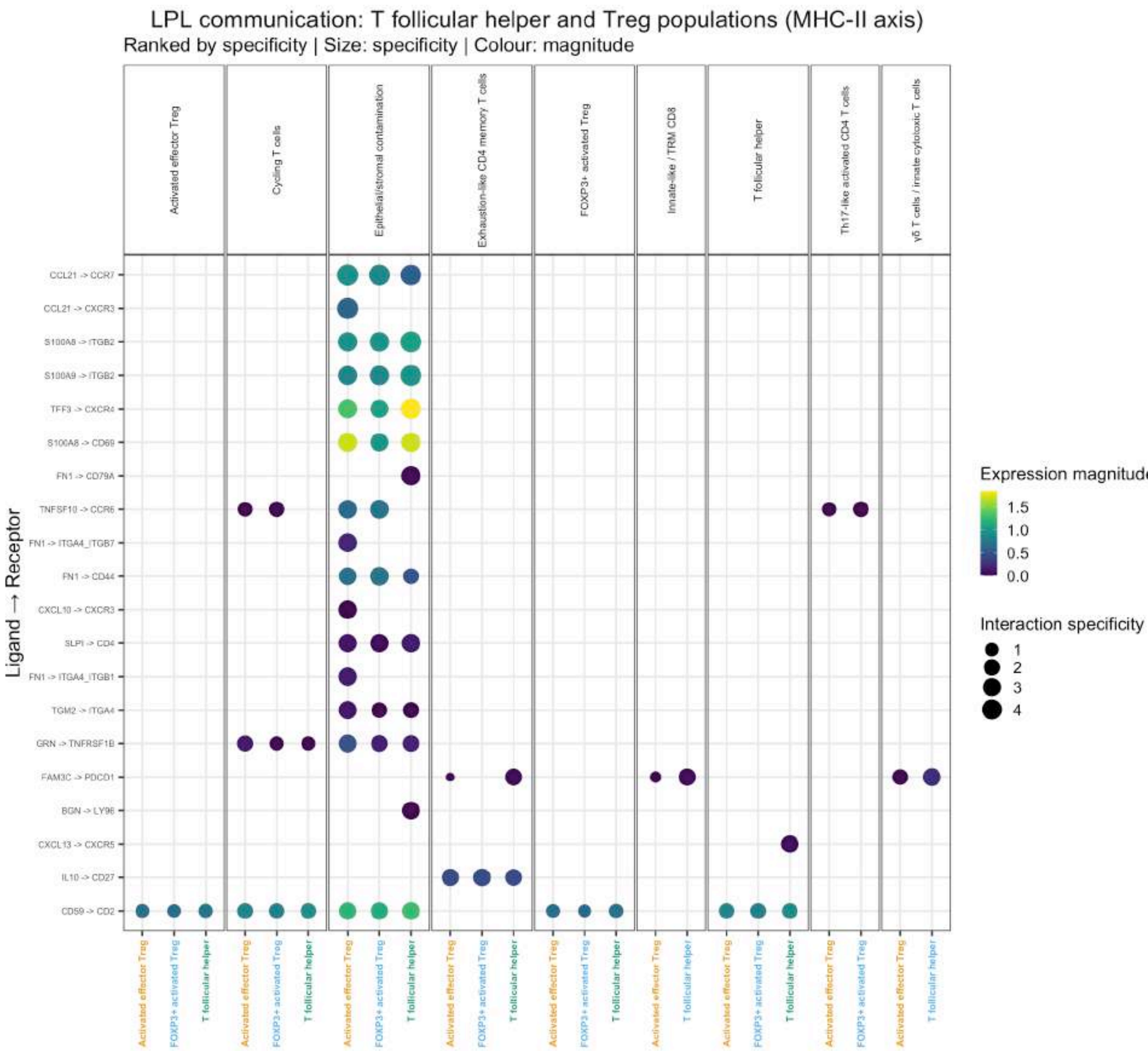


LPL communication: interactions targeting Activated effector Treg
Ranked by specificity | Size: specificity | Colour: magnitude
Activated CD4 memory T cells
Activated cytotoxic CD8 T cells
Activated effector Treg
CD4 TRM-like activated memory T cells
Cycling T cells
Epithelial/stromal contamination
Exhaustion-like CD4 memory T cells
FOXP3+ activated Treg
IFN-activated CD8 cytotoxic T cells
Innate-like / TRM CD8
NK-like cytotoxic T cells
T follicular helper
Th17-like activated CD4 T cells
TRM-like CD8 cytotoxic T cells
γδ T cells / innate cytotoxic T cells
Ligand → Receptor
CCL21 -> CCR7
CCL21 -> CXCR3
S100A8 -> ITGB2
S100A8 -> CD69
S100A9 -> ITGB2
TFF3 -> CXCR4
FN1 -> ITGA4_ITGB7
TNFSF10 -> CCR6
CXCL10 -> CXCR3
FN1 -> ITGA4_ITGB1
SLPI -> CD4
TGM2 -> ITGA4
GRN -> TNFRSF1B
FN1 -> CD44
CD59 -> CD2
IL10 -> CD27
LAMB3 -> CD44
HLA-DRB1 -> CD4
TIMP3 -> CD44
HLA-DQA1 -> CD4
Activated effector Treg
Expression magnitude
1.6
1.2
0.8
0.4
0.0
Interaction specificity
0
1
2
3
4

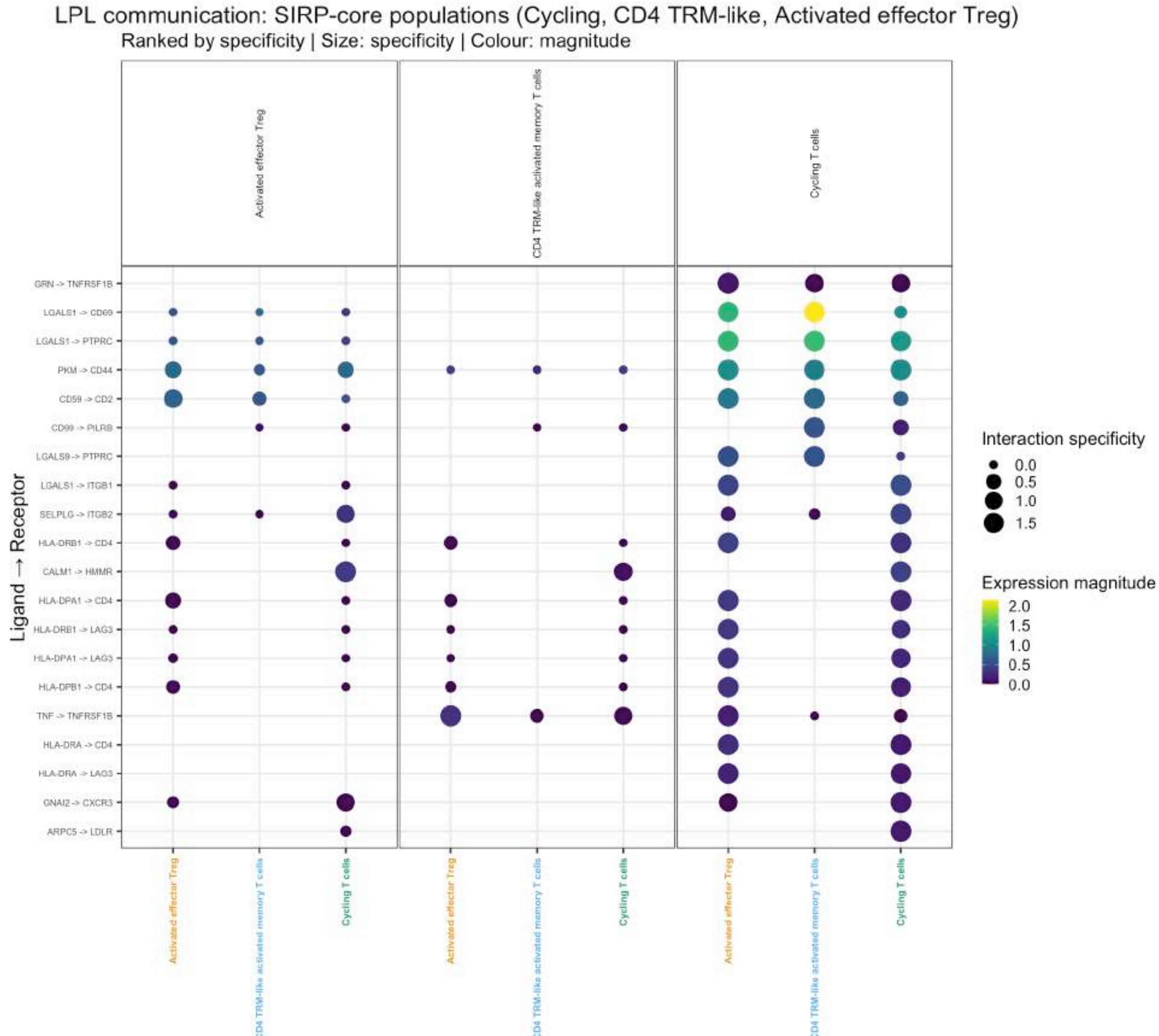
Figure 6.



LPL communication: Innate-like/TRM CD8 → IL2-axis targets
Ranked by expression magnitude | Size: specificity | Colour: magnitude
Innate-like / TRM CD8
Ligand → Receptor
B2M -> CD3D
B2M -> CD3G
HLA-B -> CD3D
HLA-A -> CD3D
HLA-C -> CD3D
B2M -> CD247
HMGB1 -> CXCR4
HLA-B -> CD3G
HLA-A -> CD3G
HLA-C -> CD3G
HLA-B -> CD8A
HLA-A -> CD8A
CD48 -> CD2
CLEC2D -> KLRB1
HLA-C -> CD8A
LGALS1 -> CD69
B2M -> KLRD1
HLA-E -> CD8A
VIM -> CD44
B2M -> TFRC
HLA-B -> CD8B
HLA-A -> CD8B
HLA-C -> CD8B
TGFB1 -> CXCR4
HLA-E -> CD8B
HLA-B -> CANX
MIF -> CD74_CXCR4
HLA-B -> KLRD1
LGALS1 -> PTPRC
MIF -> CD44_CD74
Activated effector Treg
CD4 TRM-like activated memory T cells
FOXP3+ activated Treg
T follicular helper
Expression magnitude
6
4
2
Interaction specificity
0.0
0.4
0.8
1.2
1.6



LPL communication: T follicular helper and Treg populations (MHC-II axis)
Ranked by specificity | Size: specificity | Colour: magnitude
Activated effector Treg
Cycling T cells
Epithelial/stromal contamination
Exhaustion-like CD4 memory T cells
FOXP3+ activated Treg
Innate-like / TRM CD8
T follicular helper
Th17-like activated CD4 T cells
γδ T cells / innate cytotoxic T cells
Ligand → Receptor
CCL21 -> CCR7
CCL21 -> CXCR3
S100A8 -> ITGB2
S100A9 -> ITGB2
TFF3 -> CXCR4
S100A8 -> CD69
FN1 -> CD79A
TNFSF10 -> CCR6
FN1 -> ITGA4_ITGB7
FN1 -> CD44
CXCL10 -> CXCR3
SLPI -> CD4
FN1 -> ITGA4_ITGB1
TGM2 -> ITGA4
GRN -> TNFRSF1B
FAM3C -> PDCD1
BGN -> LY96
CXCL13 -> CXCR5
IL10 -> CD27
CD59 -> CD2
Activated effector Treg
FOXP3+ activated Treg
T follicular helper
Expression magnitude
1.5
1.0
0.5
0.0
Interaction specificity
1
2
3
4



6.2 CD-vs-Control comparison (`fully_comparable_types_lpl`, 7 populations)

Combined dotplot generated with the identical `liana_bysample()` split/tier-filter schema used for IEL. Must be read with the Section 3.2 caveat on threshold divergence from CellChat's own validity table. The observed pattern (dominance of MHC-I/MHC-II/adhesion signal in both conditions, differing more in magnitude than presence/absence) is consistent with the targeted deep-dive in Section 6.1.

Figure 7.

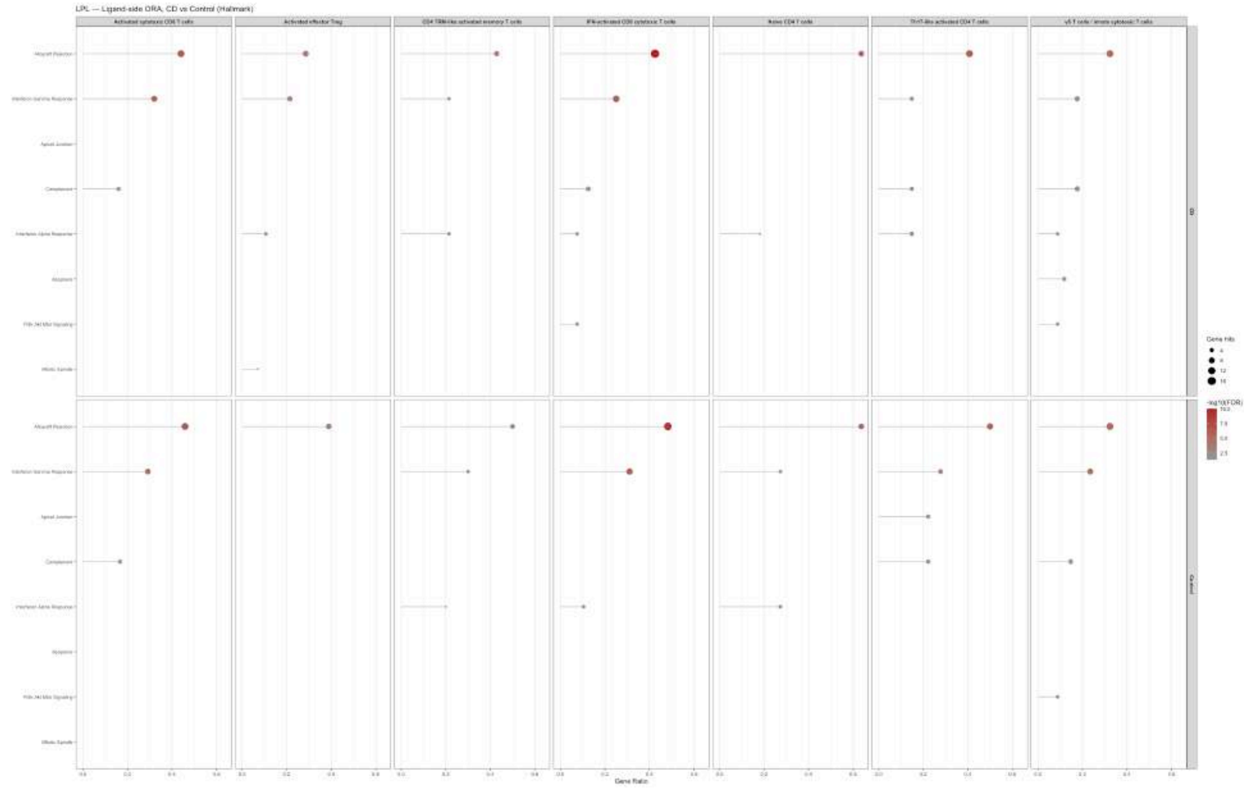
6.3 Hallmark ORA (ligand and receptor side, `fully_comparable_types_lpl`)

`ALLOGRAFT_REJECTION` dominates nearly every population/condition panel in both directions — flagged initially as a possible structural artifact of a background already restricted to immune-recognition genes, then resolved by gene-level decomposition of the intersecting hits:

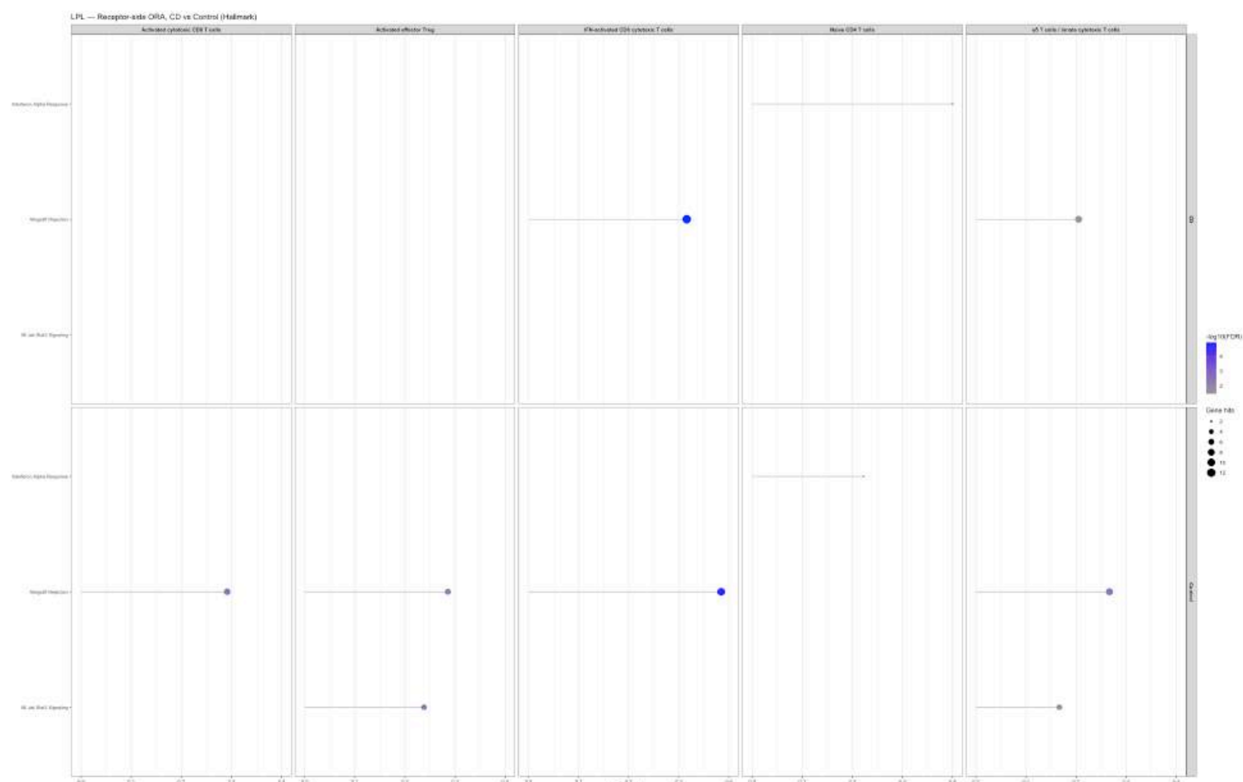
- **Generic, uninformative core, shared everywhere**: B2M, HLA-A, HLA-E, LCK — identical across Treg-CD, Naive CD4-CD, Treg-Control; not discriminating.
- **TGFB1**: present on Activated effector Treg in *both* conditions, absent on Naive CD4 T cells — consistent with Treg identity, stable rather than disease-differential; a methods-validation signal rather than a disease finding.
- **IL10**: present *only* on Activated effector Treg in Control, absent in CD — **the strongest ORA finding in LPL**, convergent with the independently observed IL10 loss in the CellChat rankNet analysis (companion report, Section 6.5/GSEA cross-validation note; originally flagged there as fragile, one interaction) — two independent sources now support attenuated anti-inflammatory tone in this population during active inflammation.
- **CD47, ITGB2, HLA-DMA**: newly present on Treg-CD vs Treg-Control — CD47 convergent with the already-confirmed SIRPG–CD47 axis (Section 6.1).
- **TNF, TIMP1**: present only on Naive CD4-CD, not Treg — population-specific differentiation, arguing against a blanket structural artifact.
- **Statistically null result, not a data gap**: the receptor-side ORA plot lacks panels for CD4 TRM-like activated memory T cells and Th17-like activated CD4 T cells; verified directly (`ora_receptor_compare |> filter(...)`) — rows exist but none clear `adj_pval < 0.05` (minimum observed 0.0654), consistent with a small `distinct_hits` denominator (8 and 13/24) rather than absence of signal.

Figure 8.

Ligand


LPL — Ligand-side ORA, CD vs Control (Hallmark)
Gene Ratio

Receptor


LPL — Receptor-side ORA, CD vs Control (Hallmark)
Gene Ratio

6.4 7. Cross-Validation with GSEA

IEL — complete. The recurring pattern shared between the three available GSEA cell types (Cycling T cells, Cytotoxic TRM-like, TH17) and the LIANA ligand-side ORA (`ALLOGRAFT_REJECTION`, `INTERFERON_GAMMA_RESPONSE`, `INTERFERON_ALPHA_RESPONSE`, `COMPLEMENT`, all CD-enriched) is treated as a plausible, expected convergence of broad gene sets rather than an independently informative one (same caution as Section 5.3). A genuine, unresolved tension is flagged rather than forced to a conclusion: `TNFA_SIGNALING_VIA_NFKB` is **negative** in CD across all three GSEA cell types, while both the LIANA ORA and the balanced CD-vs-Control dotplot show a TNF-superfamily axis **active/present** in CD at the co-expression level — plausibly reflecting two distinct biological layers (potential communication via L-R co-expression vs. measured downstream NFkB transcriptional response), stated in the manuscript as an open question, not resolved either way.

LPL — complete. GSEA (DESeq2 pseudobulk, Hallmark, `NES` positive = enriched in CD, same convention as IEL) is available for two LPL cell types: Cytotoxic TRM-like and Treg.

- **Generic/expected convergence with the LIANA ORA (Section 6.3)**: `ALLOGRAFT_REJECTION` is strongly CD-enriched in both GSEA cell types (Cytotoxic TRM-like: NES 2.0; Treg: NES 1.8), mirroring its near-ubiquitous dominance in the LIANA ligand-side ORA. Same caution applies as everywhere else this gene set appears in this report — expected given a broad, immune-centric gene set, not independently discriminating on its own. `INTERFERON_GAMMA_RESPONSE` shows the same pattern (positive NES in both GSEA cell types; also recurrent in the LIANA ORA on Activated effector Treg and several other populations) — plausible convergence, same generic-gene-set caveat.
- **The most important result of this section: a three-way convergence on IL2 being Treg-specific, contradicting CellChat's original characterization.** `IL2_STAT5_SIGNALING` is significantly CD-enriched specifically in the **Treg** transcriptome (NES 1.7) and does **not** appear among the top Hallmark hits for Cytotoxic TRM-like — i.e., GSEA independently places this transcriptional program in Treg specifically, not in a CD8/cytotoxic population. This aligns directly with LIANA's finding (Section 6.1) that the high-affinity IL2RA_IL2RB_IL2RG receptor complex is specific to both Treg populations, and with the external-literature basis for that reading (constitutive CD25/IL2RA on Treg). It contradicts CellChat's original pathway characterization (high-affinity complex assigned to T follicular helper/CD4 TRM-like, not Treg) on a second, independent axis. **Taken together, GSEA and LIANA now agree with each other and with canonical immunology; CellChat's original receptor-complex assignment is the outlier on all three counts.** This upgrades the IL2 finding from "LIANA disagrees with CellChat, literature favors LIANA" (Section 6.1/9) to a genuinely converging three-method picture, and is worth revisiting in the companion CellChat report's Section 7, which currently frames the GSEA–CellChat IL2 relationship as only a "partial, not clean" directional convergence — that framing was written before this LIANA/literature arbitration and likely undersells how one-sided the evidence now is.
- **`TNFA_SIGNALING_VIA_NFKB` is negative in CD in both LPL cell types** (Cytotoxic TRM-like NES -1.6; Treg NES -1.8) — consistent with, not in tension with, the LPL CellChat/LIANA picture: unlike IEL (Section 7, IEL), no LPL deep-dive pathway claims an *active* TNF-superfamily ligand–receptor axis in CD to create a comparable tension. The companion CellChat report already notes that canonical TNF/IFNG/IL17/TGFb signaling is undetected by CellChat in **both** compartments (Section 2.2/10, companion report) — the LPL GSEA result is therefore compatible with that absence, not contradictory to it. The IEL-specific TNF/NFkB tension (Section 7, IEL above) should **not** be generalized to LPL in the manuscript.
- **Metabolic/proliferative programs exclusive to GSEA** (Oxidative Phosphorylation — the strongest hit in both cell types by FDR; Myc Targets V1, MTORC1 Signaling, DNA Repair, Adipogenesis, Cholesterol Homeostasis) have no ligand/receptor equivalent and are not testable against LIANA or CellChat by construction — reported here for completeness, not as cross-validated findings.

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7 Known Technical Issues (LIANA) — Methods Note

Documented for reproducibility; several apply identically to both compartments, one is LPL-specific.

1. **Seurat v5 / `GetAssayData()` incompatibility** (both compartments): `as.SingleCellExperiment()` and direct `liana_wrap()` on a Seurat v5 multi-layer object fail — known issue on both the Seurat and LIANA repositories. *Fix*: manual `SingleCellExperiment` construction via `LayerData()`.
2. **Incomplete `counts` layer** (both compartments): the compartment-subset Seurat objects had `data` already joined across samples but `counts` still split, with some samples entirely missing. *Fix*: raw counts recovered from the full integrated object (`srt_obj_merge_harmony.rds`), rejoined, subset by barcode, with an explicit `colnames`/`rownames` identity check before proceeding.
3. **`liana_dotplot()` default columns point to single-method scores** (`natmi.edge_specificity`/`sca.LRscore`), not the aggregated ranks — unless `magnitude="magnitude_rank", specificity="specificity_rank", invert_*=TRUE` are explicitly set, plots reflect consensus only in point ordering, not in color/size. Corrected in all plots from the point of detection onward.
4. **Facet illegibility beyond ~5–7 cell types**, in both dotplot and chord visualizations — `source_groups`/`target_groups` restricted to 3–6 populations per side throughout; `chord_freq()` in particular does not render a legend (a `circlize` limitation, not LIANA's), compensated with text captions.
5. **`facet_grid(..., scales="free_y")` disalignment** in CD-vs-Control comparative plots — removed in favor of shared, fixed scales with pathway order fixed explicitly via `factor(..., levels=...)`.
6. **String mismatch in cell type name (LPL-specific)**: `"Innate-like/TRM CD8"` (no spaces around the slash, as typed in code) vs. `"Innate-like / TRM CD8"` (actual level, with spaces) — caused a silent zero-row filter and a downstream `Error in combine_vars(): Faceting variables must have at least one value`. Diagnosed via `sort(unique(...))` and a direct row-count check; corrected throughout.
7. **Generic top-*N* ranking is unsuitable for pathway-specific cross-validation questions (LPL-specific)**: `liana_dotplot(ntop=...)` ranks across *all* L–R pairs within the chosen source/target subset, not within a specific pathway — ubiquitous signals (MHC-I self-recognition, CD59-CD2/CD48-CD2 adhesion) systematically outrank weaker, targeted pathways (IL2: 11 CellChat interactions total; SIRP: a single L–R pair). *Resolution*: for exploratory questions ("what dominates here"), top-*N* ranking remains appropriate (IEL dotplots, Section 5; LPL Section 6.2); for pathway-specific cross-validation, direct `filter() + str_detect()` querying of `magnitude_rank`/`specificity_rank` was used instead (Section 6.1), the same pattern already used for IEL's targeted LIGHT-LTBR diagnosis.
8. **Enrichment script file-naming carryover (LPL-specific, resolved)**: `22_LIANA_LPL_enrichment.r`, adapted from the IEL script, retained six residual `"IEL"` string occurrences in `save_rds()` calls, `save_session_info()`, and the final pipeline message. Flagged during this analysis and **confirmed corrected by the analyst** — no longer a reproducibility risk.

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8 Comparison with External Literature

Finding (this analysis)	Independent literature	Concordance
MHC-I as broad, near-ubiquitous signal, both compartments	Consistent with the well-established near-universal expression of classical MHC-I on nucleated cells; not a disease-specific claim	Expected, structural
CDH1(E-cadherin)–CD103(ITGAE_ITGB7), IEL	Established retention mechanism for intraepithelial lymphocytes and CD103 Tregs (Yang et al., *Immunity* 2025) — same literature basis already cited in the CellChat companion report, Section 5.6/9	Direct mechanistic support, cross-method confirmed
IL2 receptor-complex pattern specific to Treg populations, LPL	Constitutive CD25(IL2RA) expression on Treg and the resulting high-affinity IL2RA-IL2RB-IL2RG trimeric receptor is textbook immunology, underlying the rationale for Treg-selective low-dose IL-2 and IL-2 mutein therapies in inflammatory disease	Direct, canonical — The LIANA and GSEA results provide convergent evidence supporting a Treg-centered IL-2 axis and are concordant with the established biology of the high-affinity IL-2 receptor on Tregs. This evidence argues against the original CellChat receptor assignment to Tfh/CD4-TRM-like cells
SIRPG–CD47 reorganization, LPL	SIRPG (SIRP 2) is T/NK-restricted (unlike myeloid SIRPA) and mediates T cell costimulation; SIRPG expression tracks with T cell differentiation/exhaustion state (Piccio et al., *Blood* 2005) — same literature basis as the CellChat companion report, Section 6.3/9	Mechanistically plausible, cross-method confirmed
B2M→TFRC/APLP2 as a non-canonical top signal, LPL MHC-I dotplot	B2M is an obligate subunit of HFE, which modulates TFRC affinity for transferrin (iron-homeostasis axis) — a real, curated interaction (verified via `select_resource("Consensus")`) but mechanistically distinct from MHC-I antigen presentation	Confirmed as genuine but must not be conflated with the MHC-I finding it superficially resembles

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9 Cross-Compartment Synthesis

1. **LIANA independently reproduces the two strongest, most literature-anchored CellChat findings in each compartment** — CDH1–CD103 in IEL, SIRPG–CD47 in LPL — using an alternative ligand–receptor inference framework based on multiple scoring methods and the LIANA Consensus resource. This is the strongest available form of validation in this project and should anchor the manuscript's methods section on cross-validation.

2. **LPL, not IEL, is where genuine cross-method tension emerged**, and in both cases the tension was resolved rather than left open: IL2 by external literature (LIANA's pattern is canonical; CellChat's is inverted), MHC-II by direct re-verification of the CellChat object itself (the original claim was simply wrong). No equivalent tension arose in IEL, where all four targeted CellChat findings were either confirmed or, in one case (MHC-II), found confounded by the same compositional constraint already known from CellChat.
3. **The MHC-II correction (LPL) is the clearest concrete justification for having built an orthogonal cross-method comparison at all** — it did not just add confidence to an existing claim, it caught an error in the primary analysis before manuscript submission. This should be stated explicitly and without hedging in the discussion of methodology.
4. **Cell type validity criteria differ meaningfully between CellChat and LIANA**, most consequentially in LPL (Section 3.2) — this is a genuine methodological choice difference (per-condition `min.cells` threshold vs. joint-minimum threshold), not an error in either report, but it must never be silently elided when the two reports are read together.
5. **GSEA cross-validation is now complete for both compartments** (Section 7). In LPL it delivers the single strongest cross-method result in this report: `IL2_STAT5_SIGNALING` is independently placed in the Treg transcriptome by GSEA, matching LIANA's Treg-specific receptor-complex finding and contradicting CellChat's original Tfh/CD4-TRM-like assignment — a three-method convergence (two of three, GSEA and LIANA, in agreement; the third, CellChat, is the outlier) that should be highlighted prominently in the manuscript's IL2/Treg discussion, and that also motivates revisiting the "partial, not clean" convergence language in the CellChat companion report's Section 7.

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10 Limitations

1. **Threshold divergence from CellChat's validity tables** (Section 3.2) governs how LIANA's LPL "fully comparable" claims should be read relative to CellChat's own — not interchangeable sets.
2. **Cycling T cells (LPL)**: treated as adequately represented in CellChat's SIRP deep-dive but falls below the LIANA `<30` threshold — a specific, named instance of tier disagreement between the two tools, not resolved by adopting one criterion as authoritative.
3. **Generic top-*N* dotplot ranking is unsuitable for pathway-specific cross-validation** (Section 8, item 7) — any future extension of this pipeline (additional pathways, additional compartments) should default to direct gene-level filtering for targeted validation questions from the outset, not discover this limitation empirically as this analysis did.
4. **LPL GSEA cross-validation** (Section 7) — complete as of this revision; only two LPL cell types (Cytotoxic TRM-like, Treg) have GSEA available, versus three for IEL, so compartment coverage remains asymmetric even though the cross-validation step itself is no longer missing.
5. **Enrichment script file-naming bug — confirmed fixed** by the analyst; no longer an open item (previously Section 8, item 8).
6. **Figure availability in this consolidated report is asymmetric between compartments**: LPL figures (Sections 4, 6, 7) are embedded directly from source files; IEL figures were not available during report consolidation and are marked with explicit placeholders ( ) at every relevant location rather than omitted silently. These must be inserted from the `plots_liana_iel` output directory before this report is considered visually complete.
7. **Not independently re-derived in this report**: no new LIANA computation was performed here; all quantitative values are taken from the four source documents and cross-checked internally for consistency, but not re-executed against the raw data during report consolidation.
8. **B2M→TFRC and similar non-canonical curated interactions**: their presence in the Consensus resource was verified for this one case (Section 6.1/9) but the resource was not systematically audited for other similarly misleading entries elsewhere in the pipeline — a residual risk when interpreting any unfiltered top-*N* ranking output.

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11 Reproducibility

Environment (shared with the CellChat and GSEA/pseudobulk pipelines):

Package	Notes
R	4.6.0
Seurat	5.5.0
LIANA	0.1.14 (`saezlab/liana`)
msigdbr	26.1.0

Reference scripts: - `21_LIANA_IEL.r`, `22_LIANA_IEL_enrichment.r` - `21_LIANA_LPL.r`, `22_LIANA_LPL_enrichment.r`
Sources consulted: LIANA R package documentation (`liana_tutorial.html`, `liana_cc2tensor.html`, `liana_intracell.html`), Single-cell best practices book (sc-best-practices.org, Cell–Cell Communication chapter), Seurat/LIANA issue trackers, and direct empirical verification via `select_resource()`, `subsetCommunication()` (on the CellChat object, for the MHC-II correction), and targeted `filter()`/`str_detect()` queries where dotplot ranking was insufficient (Section 8).

Random seed: `1234`, called identically in both compartments, consistent with the CellChat pipeline's convention.

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12 Deliverables

- Per-compartment LIANA aggregated objects (pooled) and condition-split objects (`liana_bysample`, CD/Control)
- Global exploration plots: `heat_freq` (both compartments), `chord_freq` (restricted subsets, both compartments)
- Targeted validation dotplots and direct tabular queries for all eight pathway validations (4 per compartment) against the CellChat deep-dive
- CD-vs-Control combined dotplots for both compartments' "fully comparable" tier
- Hallmark ORA results and lollipop plots, ligand and receptor side, both compartments
- Gene-level decomposition tables for generic dominant gene sets (`ALLOGRAFT_REJECTION`, LPL)
- Four source documents consolidated into this report (`LIANA_IEL_findings_log.md`, `LIANA_IEL_pipeline_report.md`, `LIANA_LPL_findings_log.md`, `LIANA_LPL_pipeline_report.md`)
- This consolidated report

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13 Conclusion

LIANA cross-validation of the IEL and LPL CellChat findings provides exactly the kind of orthogonal check it was designed to provide: independent confirmation of the two strongest CellChat findings in each compartment (CDH1–CD103, SIRPG–CD47), a literature-arbitrated correction of a CellChat pathway characterization (IL2, LPL), and — the single most important outcome of this entire analysis — the detection and correction of a genuine error in the primary CellChat findings log (MHC-II, LPL), traced back to and confirmed directly on the CellChat object itself rather than left as an unresolved cross-method disagreement. The explicit, LIANA-specific validity framework (Section 3) and the declared divergence from CellChat's own validity criteria in LPL are presented here as integral parts of the analytical design, not limitations discovered after the fact. With GSEA cross-validation now complete for both compartments, the IL2/Treg axis stands out as the report's clearest example of convergent, multi-method evidence overturning a single primary-method characterization — GSEA and LIANA agree, CellChat's original receptor-complex assignment does not — and deserves prominent, explicit treatment in the manuscript rather than a footnote. One item remains open before this report is visually complete: the IEL figures marked with placeholders throughout (Sections 4, 6.3 and their IEL analogues) must be inserted from the `plots_liana_iel` output directory before the combined CellChat+LIANA+GSEA narrative is finalized for submission.

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14 Appendix A: Glossary of Abbreviations

Term	Definition
CD	Crohn's disease
IEL	Intraepithelial lymphocytes
LPL	Lamina propria lymphocytes
TRM	Tissue-resident memory (T cell)
L–R	Ligand–receptor
ORA	Over-representation analysis
RRA	RobustRankAggreg
SCE	SingleCellExperiment
MHC-I / MHC-II	Major histocompatibility complex, class I / II
GSEA	Gene Set Enrichment Analysis

15 Appendix B: References

1. Jaeger N., et al. (2021). Single-cell analyses of Crohn's disease tissues reveal intestinal intraepithelial T cell heterogeneity and altered subset distributions. *Nature Communications*, 12, 1921.
2. Dimitrov D., et al. LIANA: a framework for cell–cell communication inference and consensus. `saezlab/liana`, package documentation and vignettes (`liana_tutorial.html`, `liana_cc2tensor.html`, `liana_intracell.html`).
3. Single-cell best practices book, Cell–Cell Communication chapter, sc-best-practices.org.
4. Yang et al. (2025). CD103–E-cadherin retention of epithelial-resident lymphocytes. *Immunity* — same citation as CellChat companion report, Section 5.6/9.
5. Piccio L., et al. (2005). SIRPG-CD47 in T cell costimulation. *Blood* — same citation as CellChat companion report, Section 6.3/9.
6. Constitutive CD25(IL2RA) expression and the high-affinity IL2RA-IL2RB-IL2RG trimeric receptor on regulatory T cells — canonical immunology, underlying the rationale for Treg-selective low-dose IL-2 and IL-2 mutein therapeutics (general reference; specific primary source to be added at manuscript stage).